MANIPAL INSTITUTE OF TECHNOLOGY
MANIPAL
(A constituent unit of MAHE, Manipal)

Department of Electrical & Electronics Engineering

Digital System Design Lab – III Semester

MINI PROJECT SYNOPSIS

Generation of Pulse Width Modulation (PWM) signal from duty cycle and clock input

Submitted By

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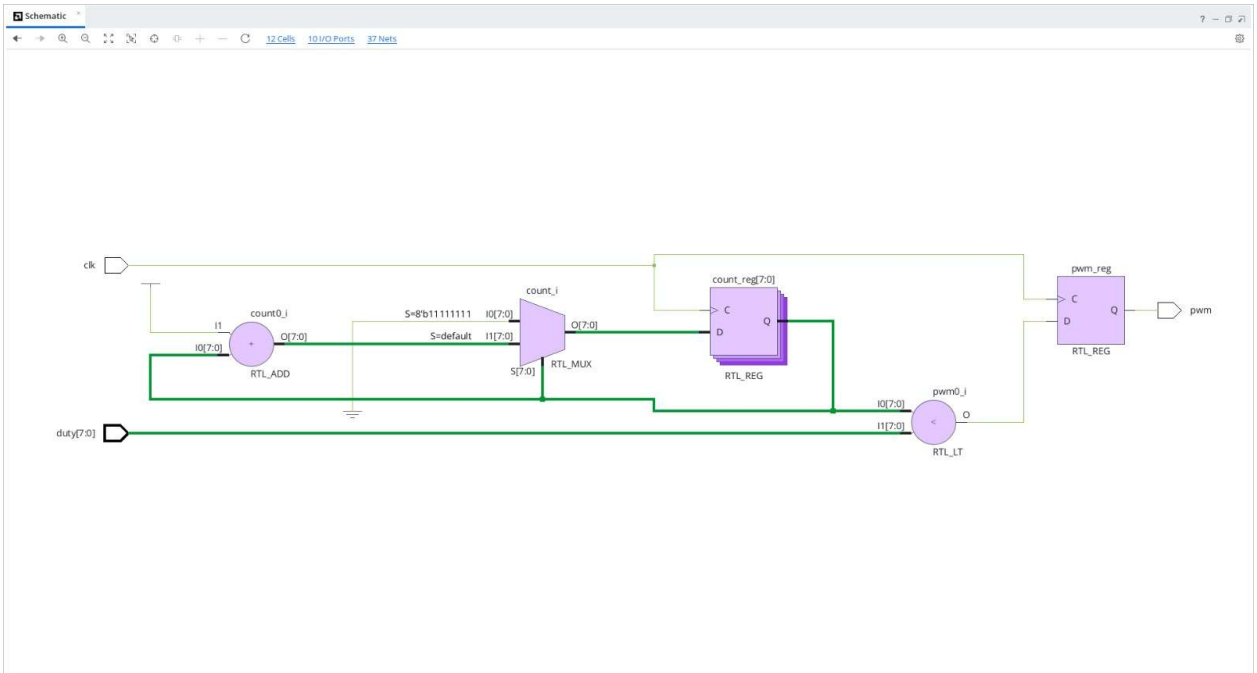
Abstract

Pulse Width Modulation (PWM) is a widely used technique for controlling the average power delivered to electronic loads by varying the width of a periodic digital pulse. This mini project focuses on the design and implementation of a 2 kHz PWM generator using Verilog HDL with behavioral modelling. The system accepts a 512 kHz input clock and an 8-bit duty-cycle input, allowing the duty cycle of the generated PWM signal to be digitally controlled.

The design uses an 8-bit counter to generate a 256-clock-cycle PWM period. The counter value is continuously compared with the 8-bit duty-cycle input to determine the state of the PWM output. A duty-cycle value of 00H produces approximately 0% duty cycle; 80H produces approximately 50% duty cycle, and FFH produces approximately 100% duty cycle. Thus, the design provides 256 possible duty-cycle levels while maintaining a PWM frequency of approximately 2 kHz.

This project is intended to demonstrate the application of digital design concepts, counters, clock-based timing, and behavioral Verilog modelling. The design can be simulated and subsequently implemented on an FPGA to observe the generated PWM waveform and verify the effect of changing the duty-cycle input.

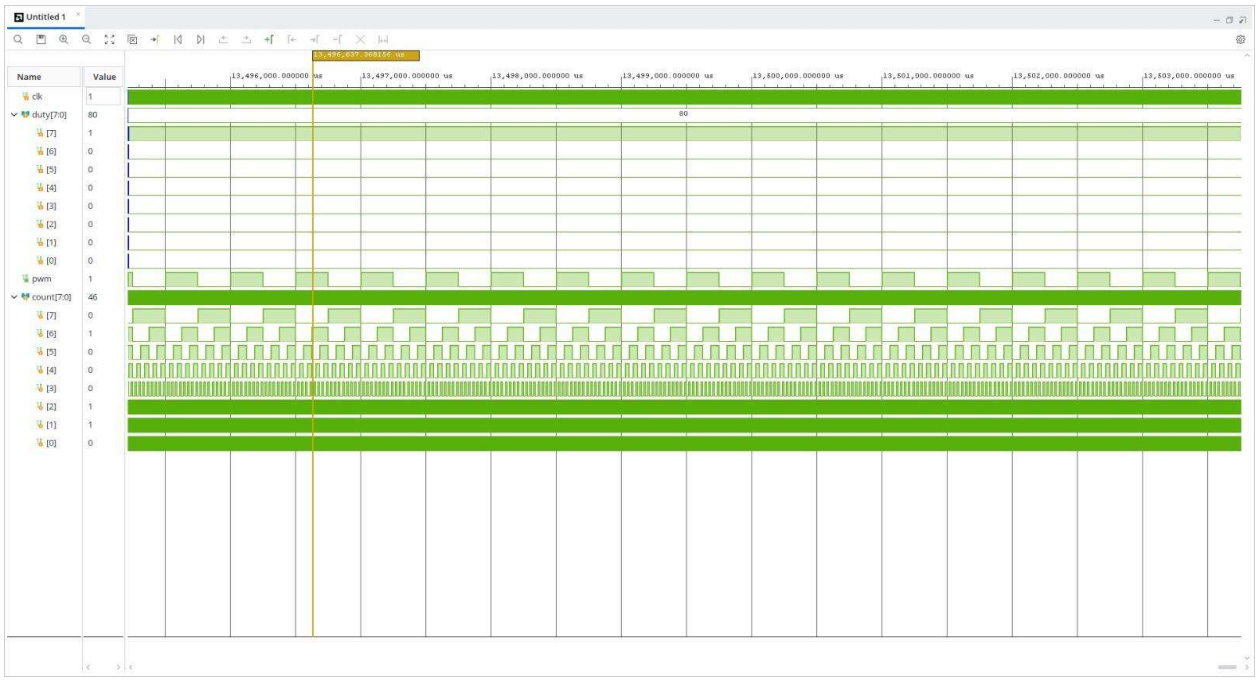
Block Diagram



Application

1. Servo and Motor control: The PWM output of a microcontroller is almost never used to directly drive a motor. Instead, this signal is interpreted by a motor controller, which converts it into a more accurate PWM signal for the motor.
2. Telecommunications: PWM is a form of signal modulation where the widths of the pulses correspond to specific data values encoded at one end and decoded at the other.
3. Power Delivery: PWM can be used to control the amount of power delivered to a load without incurring the losses that would result from linear power delivery by resistive means.
4. Audio Effects: Varying the duty cycle of a pulse waveform in a synthesis instrument creates useful timbral variations.

Simulation



Contribution

1. Soham: Simulation and Code
2. Kumar Aditya: Write-up
3. Jayendra: Verification

References

1. https://en.wikipedia.org/wiki/Pulse-width_modulation